What is NeSI?

National High Performance Computers & eResearch Services

NeSI Support: support@nesi.org.nz

REANNZ - NeSI Integration

Strengthening eResearch infrastructure through the integration of NeSI into REANNZ

Watch this space as we bring you more updates!

Growing the computational capability of New Zealand researchers for our wellbeing

collaborative values + *eResearch expertise* + *shared infrastructure*

Disciplines Supported

- Biology
- Engineering
- Astronomy
- Physics
- Chemistry
- Computer Science
- Medical Science
- Earth Science
- Social Science
- Mathematics

Core Services

- High Performance Computing & Analytics
- Consultancy
- Training
- Data Transfer & Share

Shared Infrastructure

- Batch-queue HPC systems
- Interactive computing environments
- Scalable, specialised storage
- Research Developer Cloud

genomics aotearoa

MALAG
INSTIT
OF MEDICAL R

Antarctic Science Platform

agresearch
āta mātai, mātai whetū

NATURAL HAZARDS
RESEARCH PLATFORM

AUT

RSE
RSE-AUNZ

globus

Departme
Conserv
Te Papa At

CallaghanInnova
New Zealand's Innovation A

Plant & Food™ Research
Rangahau Ahumāra Kai

THE UNIVERSITY OF WAIKATO
Te Whare Wānanga o Waikato

Science New Zealand

QuakeCoRE
NZ Centre for Earthquake Resilience
Te Hiranga Rū

hrc nz
Health Research Council of New Zealand
Te Kaunihera Rangahau Hauora o Aotearoa

BRAGATO RESEARCH INSTITUTE
NEW ZEALAND GRAPE AND WINE RESEARCH
RANGAHAU KAREPE, WĀINA O AOTEAROA

PACIFIC EDGE LTD
CANCER DIAGNOSTICS COMPANY

Hewlett Packard Enterprise

BIOTELLIGA
Harnessing nature's genius

REANN

Livestock Improvement™

MARSDEN FUND
TE PŪTEA RANGAHAU A MARSDEN

Ministry for Primary Industries
Manatū Ahu Matua

RUTHERFORD
DISCOVERY FELLOWSHIPS

ROYAL SOCIETY
TE APĀRANGI

MOANA PROJECT

MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

MetService
TE RATONGA TIRORANGI

MetOcean
SOLUTIONS

AeRO
Australasian eResearch Organisations

aarnet
Australia's Academic and Research Network

GNS SCIENCE
TE PŪ AO

UC
UNIVERSITY
CANTERBU
Te Whare Wānanga o W
CHRISTCHURCH NEW ZE

1897
VICTORIA UNIVER
WELLING
TE HERENGA

THE CARPENTRIES

Pawsey

NCI AUSTRALIA

MINISTRY OF BUSINESS, INNOVATION & EMPLOYMENT
HĪKINA WHAKATUTUKI

Waipapa Taumata Rau
University of Auckland

NIWA
Taihoro Nukurangi

UNIVERSITY of OTAGO
Te Whare Wānanga o Otāgo
NEW ZEALAND
SAPERE AUDE

Manaaki Whenua
Landcare Research

HPC & data analytics

Data services

Training

Consultancy

NeSI
New Zealand eScience Infrastructure

NeSI's Core Services

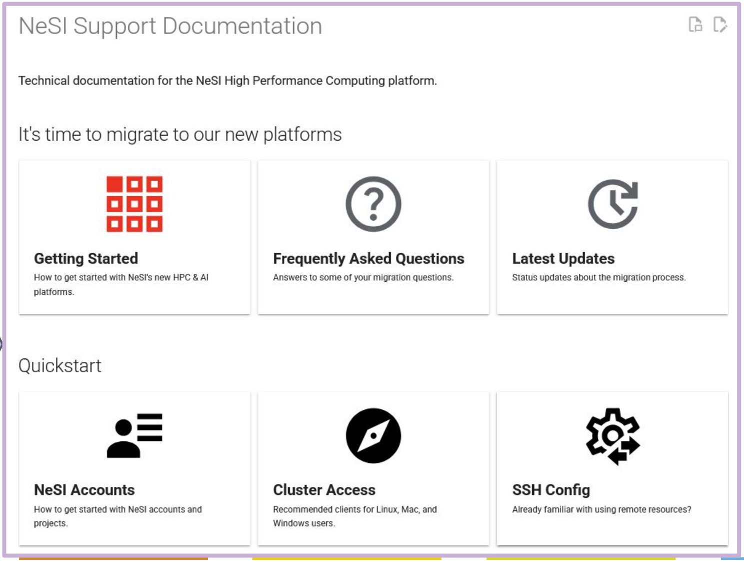
High Performance Computing & Data Analytics

NeSI gives you access to:

- Powerful high performance computers and
- A dedicated apps support team to answer your questions

There is no question too big or too small.
Please reach out to support@nesi.org.nz anytime, or check out our support website at https://docs.nesi.org.nz/

New HPC being set up in Auckland, NZ



What is High Performance Computing (HPC)?

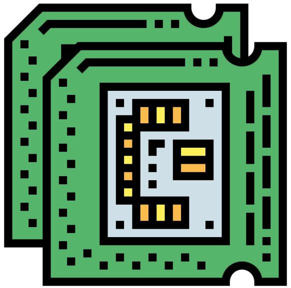
Personal laptop =
a few cores and GBs
of memory

HPC =
thousands cores and
TBs of memory

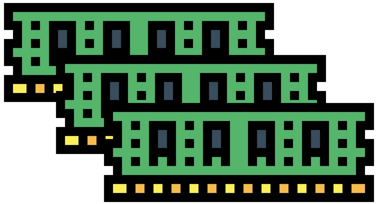
Contact support@nesi.org.nz

Reasons to use HPC

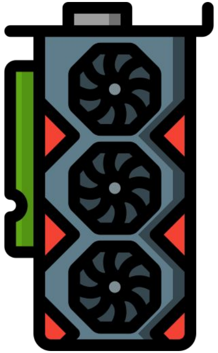
Your problem can use **multiple CPUs**



Your problem needs **more memory** (RAM)

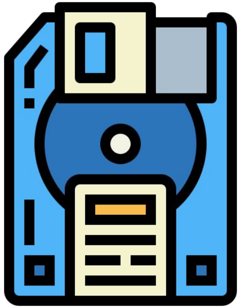


Your problem can use **specialised resources** (GPUs)



Your problem needs a **long time to run** (days/weeks)

Your problem works on data too large for **available storage**



Icons by smalllikeart

NeSI HPC

Compute:

- 19,368 cores
- 92,256 GB RAM
- 24 A100 GPUs
- 8 H100 GPUs
- 16 L4 GPUs

Storage:

- 3 PB of storage with seamless data movement to higher performance storage for running jobs.
- 50 TB cache, with 2 PB of tapes

Intel Cascade

- 2 nodes with 40 cores and 1500 GB RAM per node
- 1 node with 88 cores and 6000 GB RAM

AMD Milan

- 56 nodes with 128 cores and 512 GB RAM per node
- 8 nodes with 128 cores and 512 GB RAM per node
- 4 A100 GPU nodes with 4 GPUs, 64 cores and 512 GB RAM per node

AMD Genoa

- 44 nodes with 168 cores and 384 GB RAM per node
- 8 nodes with 168 cores ad 1536 GB RAM per node
- 4 A100 GPU nodes with 2 GPUs, 168 cores and 718 GB RAM per node
- 4 H100 GPU nodes with 2 GPUs, 168 cores and 1536 GB RAM per node
- 4 L4 GPU nodes with 4 GPUs, 168 cores and 1536 GB RAM per node

Storage

- 443 TB NVMe high performance cache, auto tiering to 3 PB Ceph Object storage
- 50 TB cache presented via s3, with 2 PB initial capacity of LT09 tapes

Data Services

National Data Transfer Platform

Secure, high-speed transfer & share capabilities for large research datasets.

REANNZ
globus

Globus endpoints in Aotearoa NZ:

NeSI New Zealand eScience Infrastructure
NIWA Taihoro Nukurangi
University of Auckland Waipapa Taumata Rau New Zealand
University of Otago Te Whare Wānanga o Otāgo New Zealand
Manaaki Whenua Landcare Research
agresearch āta mātai, mātai whetū
Plant & Food™ Research Rangahau Ahumāra Kai
scion FORESTS ▪ PRODUCTS ▪ INNOVATION

National Data Transfer Platform

Shared Datasets

High Performance Storage
7 PB

Long-term Storage
4 PB

Backup Storage
4 PB

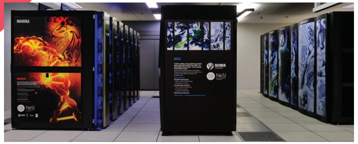
Shared Datasets

Virtual labs and portals.
Support of FAIR principles.

genomics aotearoa

Co-designing longer-term data management solutions:

- Aotearoa Genomic Data Repository
- Rakeiora project



Data Storage

Resources for projects using NeSI's HPC platform.

Training

A service available to all New Zealand-based researchers, providing training and support to build capability & develop skills in high performance computing, data science, and digital research methods.

How We Support Digital Skills and Capability Developmer

📺 **Webinars & Intro Sessions:** Regular tips, tools & HPC starter sessions

🛠️ **Workshops:** Hands-on code profiling & optimisation

🧬 **Bioinformatics Training:** National collaboration with Genomics Aotearoa

🎓 **Instructor Training:** National support via Carpentries platinum membership

👩‍🏫 **Carpentries Workshops:**Nationwide workshops by certified instructors

▶️ **YouTube Library**: Dynamic, updated online training hub

📚 **Training Catalogue**: Open-source materials on website & GitHub

☁️ **Cloud-Based Training Environment:** On-demand HPC & tailored data science access

🕒 **Office Hours**: Weekly online drop-in for user support

To stay 'in the loop':

- Subscribe to the NeSI training mailing list on: https://www.nesi.org.nz/services/training
- Follow us on LinkedIn
- Learn about more upcoming events: https://www.nesi.org.nz/community/events

Computational Science consultancy

- A service offered to NeSI platform users, generally at no cost to the researcher
- NeSI Research Software Engineers work directly with research group members
- Goal is to **raise the capability** of the research group

Our Research Software Engineers can assist with:

- **Workflow parallelisation** – allowing more inputs to be processed simultaneously
- **Software parallelisation** – use of technologies such as OpenMP or MPI to process one single input more quickly
- **Code optimisation** – redesign of algorithms to improve overall speed or efficiency of resource use
- **Improving I/O performance** – speed up reading from or writing to the disk, or to reduce the amount of data that must be read or written
- **Porting to GPU** – accelerate code by offloading computations to a coprocessor
- **Improving software sustainability** – introducing best practices such as version control and unit testing

Contact support@nesi.org.nz

NeSI
New Zealand eScience Infrastructure

What you need to know

Environment

Linux	*No windows, sorry*
Command Line*	*Using Bash Unix Shell* *- Check out our command cheats sheets*
Batch Processing*	*Can run without your direction*
Scheduled*	*Resources booked though SLURM*
Shared Resource*	*Be considerate :)*

** Mostly*

Software

Many software packages are **already installed & maintained**

List can be found @ support.nesi.org.nz under:
‘Supported Applications’

We may centrally **install software on request**

Provided it has a Linux version

Licensed software can be used on the platforms, though NeSI does not usual maintain licenses

You may also install **your own software**

Data Security

- We offer secure, NZ based data storage
- Data is by default only accessible by project team members and NeSI support
- Only the project owner, or designated project team members, can add users to a project
- Secure, high speed data transfer with Globus
- Daily data snapshots, preserved for 7 days
- Offsite backups
- If you have unique data security requirements, or have any questions or concerns, please contact us

How to access NeSI

Access Eligibility

- **Collaborator:**
  Collaborator institutions are eligible to access the platforms through the Collaborator pool (University of Auckland, University of Otago, Manaaki Whenua - Landcare Research, and NIWA).

- **Postgraduates:**
  Postgraduate students from NZ tertiary institutions are eligible.

- **Merit:**
  Users with a peer-reviewed and contestable national grant or contract.

- **Subscription:**
  Alternatively you or your institution/group can purchase a subscription

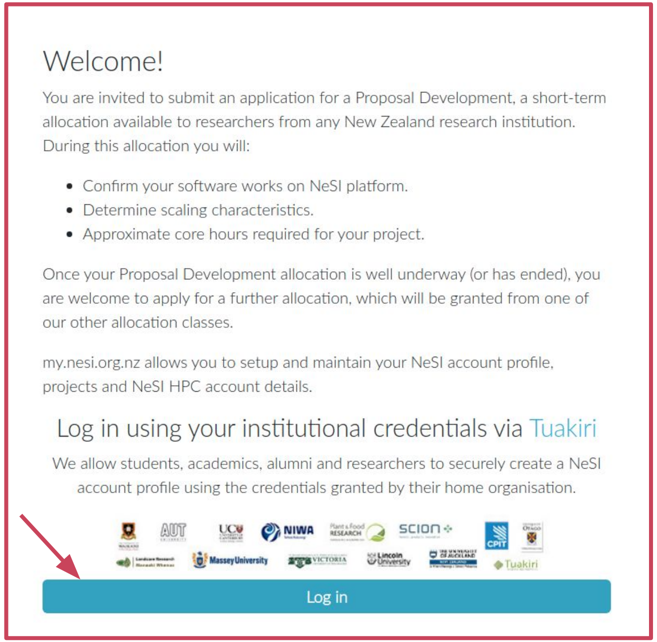
- **Proposal Development:**
  A small, free, test allocation to try out the NeSI platforms.

Contact us at support@nesi.org.nz if you have any questions.

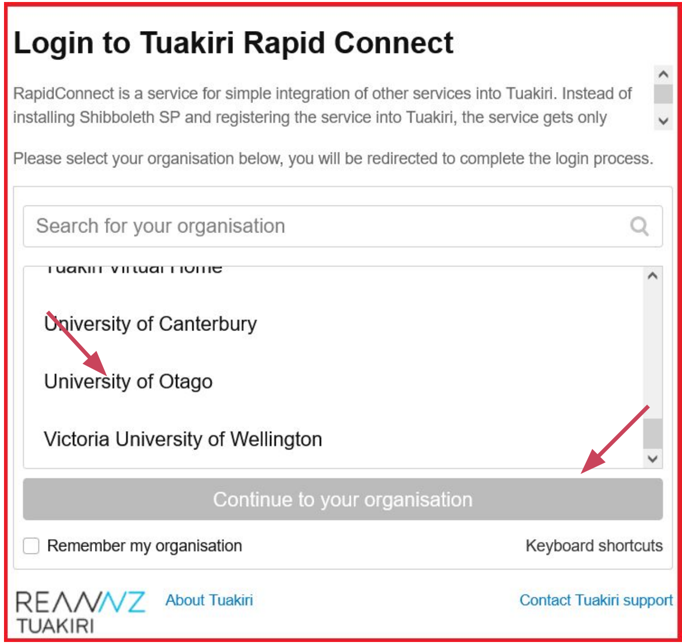
Requesting an Account

1. https://my.nesi.org.nz/

2.



3.



Note: Users from non-Tuakiri federated institutions need to sign up here:
https://my.nesi.org.nz/register

Accessing the platforms

Mac/Linux users through terminal

Windows have to download a client
(e.g. WSL, MobaXterm, etc)

Using JupyterHub, via browser

OPEN OnDemand

Details on setting up your login configuration can be found at support.nesi.org.nz

NeSI OnDemand

NeSI OnDemand allows you to access the NeSI HPC resources through the browser as an alternative SSH terminal, while also offering interactive applications for you to work with. The applications currently include:

- JupyterLab
- VSCode
- RStudio
- A virtual desktop

- To access NeSI OnDemand merely navigate to https://ondemand.nesi.org.nz/ with your browser and log in with your institutional credentials.
- You can find more in depth instructions for accessing NeSI OnDemand on our support pages.

Research Developer Cloud

https://www.nesi.org.nz/developercloud

Cloud environment for developing custom solutions

- **Cloud native support** – spin up VMs to build custom research solutions on our cloud
- **Infrastructure as Code** - APIs to support as-code approach for transparency, maintainability, and repeatability
- **Collaborate with NeSI experts** – collaborate with us to build shared knowledge for the sector

Our platform's cloud building blocks include:

Compute
Virtual machines optimised for high-performance computational needs. Multiple flavours of CPU and GPU resources to support a range of compute and memory requirements.

Images
Tailored operating systems to meet your research computing and data needs. Ready-to-use options available, as well as capability to create custom images and contribute to a pool of community-developed images.

Storage
Scalable storage space that can be dynamically mounted to your Compute instances. Options to encrypt storage volumes for added security.

Networks
Fast, reliable, and secure connectivity built on the REANNZ national network. Options for network customisation and security groups.

Identity
Identity management services to create application credentials and control access to your projects.

Application Programming Interface (API)
All services are programmable via a public API to enable repeatable definition of infrastructure through software code.

Let's stay connected

Interested in news & events ...

Join our mailing list at https://www.nesi.org.nz/
(training alerts, newsletters, event announcements, etc.)

Follow us on LinkedIn
New Zealand eScience Infrastructure

Technical questions ...

Email our Team: support@nesi.org.nz

Visit our Support site: https://docs.nesi.org.nz/

Weekly online Office Hours: https://docs.nesi.org.nz/Getting_Started/Getting_Help/Weekly_Online_Office_Hours/

Ready to get started ...

Apply for access: https://my.nesi.org.nz/

Sign up for Intro To HPC With NeSI this Friday…

https://www.eventbrite.co.nz/e/introduction-to-high-performance-computing-with-nesi-tickets-1341408903139